Campus Parking and Exit Carpool Coordination System

Julián David Muñoz Revelo Nicolás Acero Ladino
Santiago Alexander García Bermeo Miguel Mateo Guillén Guzmán

Systems Engineering Program
Universidad Distrital Francisco José de Caldas

Semester 2026-I

Project Evolution and Integration (Workshops 1-4)

The project lifecycle demonstrates a comprehensive systems engineering journey, where each phase naturally and rigorously built into the next:

- **Workshop 1 – Systems Analysis:** Field-based empirical data collection, quantitative surveys, and identification of the initial structural disequilibrium at the Calle 40 campus.
- **Workshop 2 – System Design:** Application of **Abstraction** to model human interactions without falling into premature technological biases.
- **Workshop 3 – Robust Design and Management:** Project boundary refinement based on realistic environmental constraints and compliance analysis under international risk standards (ISO 31000).
- **Workshop 4 – Dynamic Simulation:** Computational and conceptual validation of systemic stability and human feedback loops under operational uncertainty.

Problem Context: Restrictions and Shift in Scope

Mobility across the engineering faculty is a highly complex socio-technical challenge. The original boundary conditions suffered a critical redefinition due to institutional realities:

Environmental Boundaries & Shifts:

- Since the return to the **ECCI Campus** was definitively canceled, the project's core objective was modified to focus explicitly on optimizing the Calle 40 and Calle 34 campuses.
- The systemic issue crystallized into severe user symptoms: long walking times and highly elevated external parking fees.

Symptoms & Disequilibrium:

- Fixed physical basement capacity completely saturated daily before 08:00 a.m.
- Severe **Entropy** (operational chaos) emerging due to manual, unstructured coordination at the entry access point.

Stakeholder Analysis: Alignment and Systemic Interactions

Stakeholder Group	Role in the System	Key Needs	Friction / Mitigation
Students	Dynamic system elements (Drivers/Passengers).	Reliable trip coordination and commuting cost optimization.	Inertial usage of informal chats. *Mitigation:* Parking priority incentives.
Security Guards	Operational boundary controllers in the physical space.	Real-time occupancy state visibility across physical stalls.	Operational overload. *Mitigation:* Unified validation workflows.
Campus Admin	Normative control authority of the institutional **Supersystem**.	Entry event traceability and mutual responsibility schemas.	Resistance to process change. *Mitigation:* Centralized operational dashboard.
Student Welfare	Strategic institutional sponsor for quality of life.	Insights on community well-being and collective mobility metrics.	Isolated administrative priorities. *Mitigation:* Automated usage metrics.

Data Collection & Empirical Findings (Workshop 1)

The parameters governing this open system are firmly grounded in the structured survey conducted with the university community ($N = 51$), validating real actor behavior:

- **Demographic Dominance:** 98% of recurring parking facility users are students, requiring an interface aligned with institutional identity.
- **Morning Peak Pressure:** 54.9% of respondents access the campus prior to 08:00 a.m., validating the critical bottleneck period at the entrance.
- **Readiness for Change:** 34.3% expressed an explicit willingness to share their exit journeys if provided with a centralized, high-trust mechanism.
- **Socio-Technical Maturity:** Public transit remains the baseline (60.8%), but informal carpooling already operates organically in a fragmented state. The system formalizes this practice.

Physical Parking Capacity Analysis

Structural analysis of the Calle 40 parking environment reveals a drastic imbalance between fixed physical supply and user demand:

- **Strict Physical Capacity:** The basement operates under a hard limit of **motorcycle stalls**, managed via localized security intuition.
- **Field-Measured Deficit:** Field observations documented a chronic daily overflow of approximately **90 motorcycles** spilled onto public perimeters and streets.
- **State Saturation:** Under peak morning arrival windows, the theoretical occupancy index exceeds 160%, colapsing physical check-in and triggering traffic queues on Calle 40.

Scope Refinement and Operational Feasibility

To guarantee project stability under resource constraints, the scope boundary was strategically refined between Workshop 2 and Workshop 3:

Deferred Components

- **Campus Bike-Sharing Scheme:** Postponed due to high urban perimeter road safety risks and a severe lack of local municipal cycling lanes.
- **Rigid Inter-Campus Routing:** Lowered in priority following shifting geographic and administrative contexts for this period.

Feasible Systemic Focus

- **Internal Capacity Management:** Coordinating physical stall rotation utilizing occupancy-driven allocation logic.
- **Exit Carpool Coordination:** Mitigating outbound traffic pressure by organizing student return flows toward key urban transit hubs.

Systems Principles Applied to Conceptual Design

The structural design abandons rigid software engineering frameworks, anchoring itself instead on fundamental **General Systems Theory** properties:

- **Organizational Modularity:** The design decouples the physical assignment subsystem (basement inventory) from the dynamic sharing subsystem (exit carpooling), allowing independent evolution.
- **Homeostatic Balancing:** The core system objective is to establish self-regulating feedback loops to maintain stable available parking space levels despite high daily demand volatility.
- **Holistic Perspective:** Campus mobility is treated as an indivisible, integrated whole; the collective behavior of students directly limits or alters the operational capacity of security personnel.

Dynamic Simulation Model (Workshop 4)

Long-term socio-technical system behaviors were evaluated using a customized Python stochastic model simulating interactions between physical limits and user choice profiles:

Stochastic Input Parameters

- **Active Student Agent Population:** 500 virtual agents initialized with randomized class distributions and distinct decision paths.
- **Physical Infrastructure Capacity:** A bounded limit of 110 real motorcycle stalls serving as a system restriction.
- **Environmental Stochastics:** 35

Coordination Logic: Regulating Cohesive Open Demand

Utilizing **Cybernetics** principles (control through feedback channels), the model processes open-system demand variables during each time slice:

1. **Nominal Demand Generation:** Agents initialize daily transit intentions mapped to current academic schedules.
2. **Trust and Adoption Evaluation:** The system estimates the proportion of students choosing collective routing based on historical peer reliability.
3. **Synergistic Reduction:** Every successfully validated match combines solo commutes into a single vehicular unit, mathematically reducing pressure before reaching the gate.
4. **Inventory Equilibrium:** The net effective demand hits the 110 stalls, outputting dynamic overflow rates and remaining waiting times.

Simulation Stress-Testing Scenarios

To validate the conceptual resilience of the system architecture under volatile states, three operating environments were simulated:

- **Scenario 1 – Baseline (Status Quo):** Uncoordinated traditional operation. Information isolation, stalls locked for extended periods, and maximum external street overflow.
- **Scenario 2 – Active Optimization:** Activation of entry allocation rules prioritizing high-occupancy vehicles alongside structured outbound node coordination.
- **Scenario 3 – Failure Mode (Systemic Stress):** Environment characterized by high driver cancellation rates, reduced peer-to-peer trust, and continuous adverse weather.

Simulation Results and Performance Metrics

Computational simulation runs verify that orchestrating systemic information yields benefits equivalent to expanding material assets:

Performance Indicator	Status	Optimized	Failure
Daily External Overflow	$\sim$90 vehicles/day	**<15 vehicles/day**	$\sim$55 vehicles/day
Carpool Match Efficiency	0% (Fragmented)	**78% Efficiency**	31% due cancellations
Access Gate Waiting Time	8.5 minutes avg.	**<1.8 minutes**	4.2 minutes avg.
Physical Stall Utilization	Inefficient Saturation	**92% Optimized**	Blocked

Systemic Foundation: Informational synergy reallocates demand load without needing brick-and-mortar structural expansions.

Operational Risk Management and Resilience (ISO 31000)

Following structured risk management guidelines, critical socio-technical threats are addressed through explicit balancing loops embedded in the design:

- **Inertia toward Informal Platforms (Low Adoption):**
  Mitigation: Linking physical privileges to coordination. Drivers maintaining high occupancy scores gain automatic priority booking tokens for the Calle 40 basement.
- **Disruptive Impact of Passenger/Driver "No-Shows":**
  Mitigation: Penalty feedback loops. Three consecutive trip cancellations without valid justification temporarily restrict the agent's system priority privileges.
- **Safety Risks via Physical Entry Bottlenecks:**
  Mitigation: Under localized contingencies or overflow states, the system releases priority exit slots, preventing parked vehicles from obstructing campus evacuation paths.

Systemic Mapping: Integrating Core Systems Frameworks

The global behavior of this design model is governed by explicit principles of complex systems theory:

- **Systemic Emergence:** The mitigation of physical street gridlock outside Calle 40 arises as an emergent property of student cooperation, rather than via authoritarian administrative enforcement.
- **Information Synergy:** Synthesizing the static physical layout with dynamic carpool scheduling creates a total operating throughput significantly greater than both elements working in isolation.
- **Operational Equifinality:** Recognizing that a balanced parking state can be reached via multiple distinct paths (altered class times, remote learning, or carpooling). Our design selects carpooling as the route of lowest institutional friction.

Systems Analysis Conclusions

The formulation and modeling of this system consolidate core tenets of system design applied to complex campus operations:

- **Socio-Technical Supremacy:** Logistics management systems fail if they treat technology as an isolated solution while ignoring human incentives and daily field operator realities.
- **Model-Based Validation:** Stochastic simulation provided quantitative proof that information control and regulation are powerful management alternatives when facing physical resource scarcity.
- **Methodological Traceability:** The course workflow enabled direct, traceable mapping from empirical observations (Workshop 1) to structural design abstractions (Workshop 2) and dynamic validations (Workshop 4).

Conceptual Completion and Implementation Horizon

With the completion of this methodological lifecycle, the theoretical and structural design of the system stands fully consolidated and validated under a systems thinking lens.

Readiness for the Technological Phase

By establishing an interaction matrix insulated against entropy, with explicit control limits and an adaptive incentive architecture, the project is **fully prepared for the transition into software development**.

This solid theoretical foundation ensures that subsequent technical execution (database instantiation, API construction, and matching algorithms) will build upon a highly mature logical framework, guaranteeing a high-impact, professional deployment.

Thank you for your attention!

Campus Parking and Exit Carpool Coordination System

Questions or Comments?